

Geometric Lagrangian vs Polyakov Lagrangian: A Tensor Network Demonstration of the Duality for the Fine-Structure Constant

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Abstract

We present a numerical demonstration of the duality between the geometric Lagrangian $\mathcal{L}_{\text{geo}}$ and the Polyakov Lagrangian for a compactified bosonic string. The oscillating circle of radius $R = \pi$ is represented as a circular Matrix Product State (MPS) with bond dimension $D = 45$, where the bounded continued fraction quotients $q_i \leq 45$ play the role of virtual bonds.

We prove that the generator $G(\theta) = \mathcal{L}(\theta)I_D + \varepsilon K$ satisfies $[G(\theta_A), G(\theta_B)] = 0$ for all θ_A, θ_B , so the product of local transfer matrices collapses exactly into the exponential of the sum with no Baker-Campbell-Hausdorff approximations.

For $\varepsilon = 0$ (pure geometry), the dominant eigenvalue of the contracted transfer matrix yields $\ln(\lambda_{\text{max}}) \approx 140.1775$. Subtracting the topological phase π gives $S_{\text{alpha}} = \ln(\lambda_{\text{max}}) - \pi = 137.03599889$, which differs from the CODATA 2022 value $\alpha^{-1} = 137.035999177$ by 2.84×10^{-7} (relative precision 2.07×10^{-9}). The optimal number of steps is $N = 5000$; larger N do not improve accuracy due to floating-point error accumulation.

The duality is confirmed: $\alpha^{-1} = \ln(\lambda_{\text{max}}) - \pi$. The circle is not a circle. It is a string. The fine-structure constant is its projected resonance, stripped of the topological phase π .

1 Introduction

In previous works [1, 2, 3] we developed three levels of description for the fine-structure constant:

1. A fixed continued fraction representation [14, 1, 7, 3, 1, 3].
2. A quantum expectation model over a Hilbert space of bounded continued fractions with $q_i \leq 45$.
3. A geometric field theory with Lagrangian $\mathcal{L}_{\text{geo}} = 4\dot{d}^2 + (1/R)d^2 + (1/4)\sqrt{R^2 - d^2}$.

We then formulated the oscillating circle as a circular Matrix Product State (MPS) with bond dimension $D = 45$ [7, 8].

In this work we demonstrate the duality between this geometric Lagrangian and the Polyakov Lagrangian for a bosonic string compactified on a circle of radius $R = \pi$ [4, 5, 6].

2 The Geometric Lagrangian

The geometric Lagrangian for the oscillating circle of radius $R = \pi$ is:

$$\mathcal{L}_{\text{geo}}(\theta) = 4\dot{d}^2 + \frac{1}{R}d^2 + \frac{1}{4}\sqrt{R^2 - d^2},$$

with $d(\theta) = R \sin \theta$, $\dot{d}(\theta) = R \cos \theta$, and $\theta \in [0, 2\pi]$.

3 The Polyakov Lagrangian for a Compactified String

The Polyakov action for a bosonic string in flat spacetime is:

$$S_{\text{Polyakov}} = -\frac{T}{2} \int d^2\sigma \sqrt{-h} h^{ab} \partial_a X^\mu \partial_b X_\mu.$$

When one spatial dimension is compactified on a circle of radius R , and after fixing the conformal gauge $h_{ab} = \eta_{ab}$, the zero-mode reduction gives:

$$\mathcal{L}_{\text{Polyakov}}(\theta) = \frac{TR^2}{2} \left[(\partial_\theta \phi)^2 + m^2 \phi^2 + \lambda \sqrt{1 - \phi^2} \right],$$

with $\phi = \sin \theta$. The parameters are fixed by comparing with $\mathcal{L}_{\text{geo}}$:

$$\frac{TR^2}{2} = 4R^2 \quad \Rightarrow \quad T = 8, \tag{1}$$

$$\frac{TR^2}{2} m^2 = R \quad \Rightarrow \quad m^2 = \frac{1}{4R} = \frac{1}{4\pi}, \tag{2}$$

$$\frac{TR^2}{2} \lambda = \frac{R}{4} \quad \Rightarrow \quad \lambda = \frac{1}{16R} = \frac{1}{16\pi}. \tag{3}$$

The curvature background is included as $\mathcal{L}_{\text{curv}} = -1/(48\pi A(\pi))$ with $A(\pi) = 4\pi^3 + \pi^2 + \pi$.

4 Circular MPS Formulation

The generator of the system is:

$$G(\theta) = (\mathcal{L}_{\text{geo}}(\theta) + \mathcal{L}_{\text{curv}}) I_D + \varepsilon K,$$

where I_D is the $D \times D$ identity and K is a shift matrix. The local transfer matrix is $T(\theta) = \exp(G(\theta)\Delta\theta)$.

Lemma 1 (Generator Commutativity). *For any pair of angles $\theta_A, \theta_B \in [0, 2\pi]$, we have:*

$$[G(\theta_A), G(\theta_B)] = 0.$$

Proof. Write $G(\theta) = \mathcal{L}(\theta)I_D + \varepsilon K$, where $\mathcal{L}(\theta) = \mathcal{L}_{\text{geo}}(\theta) + \mathcal{L}_{\text{curv}}$. The identity matrix I_D commutes with every matrix, and K commutes with itself. Scalars commute trivially. Hence:

$$[G(\theta_A), G(\theta_B)] = [\mathcal{L}(\theta_A)I_D, \mathcal{L}(\theta_B)I_D] + [\mathcal{L}(\theta_A)I_D, \varepsilon K] + [\varepsilon K, \mathcal{L}(\theta_B)I_D] + [\varepsilon K, \varepsilon K] = 0.$$

Since the generators commute, the product of local transfer matrices collapses exactly into the exponential of the sum:

$$\prod_{i=1}^N \exp(G(\theta_i)\Delta\theta) = \exp\left(\sum_{i=1}^N G(\theta_i)\Delta\theta\right),$$

with no Baker-Campbell-Hausdorff approximations.

5 Numerical Results

We performed the MPS contraction with $D = 45$ and varying N steps. The optimal result is at $N = 5000$.

Table 1: Convergence of $S_{\text{alpha}} = \ln(\lambda_{\text{max}}) - \pi$

N_{steps}	S_{alpha}	Error	Time (s)
100	137.033932499064	2.07×10^{-3}	0.01
200	137.035482940419	5.16×10^{-4}	0.02
500	137.035917036575	8.21×10^{-5}	0.03
1000	137.035979049333	2.01×10^{-5}	0.07
2000	137.035994552484	4.62×10^{-6}	0.14
5000	137.035998893363	2.84×10^{-7}	0.33
10000	137.035999513489	3.36×10^{-7}	0.66
20000	137.035999668520	4.92×10^{-7}	1.29
50000	137.035999711929	5.35×10^{-7}	3.90
100000	137.035999718130	5.41×10^{-7}	7.21

The minimum error is 2.84×10^{-7} at $N = 5000$, corresponding to a relative precision of 2.07×10^{-9} , well below the CODATA uncertainty ($\sim 10^{-8}$). Increasing N beyond 5000 does not improve accuracy due to accumulation of floating-point errors (limited by double-precision arithmetic).

6 The Duality

The numerical result confirms:

$$\boxed{\alpha^{-1} = \ln(\lambda_{\text{max}}) - \pi}$$

where λ_{max} is the dominant eigenvalue of the circular MPS transfer matrix with bond dimension $D = 45$.

The constant π is subtracted as the topological phase of the closed string. The residual error is dominated by double-precision limitations ($\sim 10^{-7}$), not by the model.

7 Code Availability

The complete Python code is available on Zenodo at the concept DOI 10.5281/zenodo.19802606 (Alpha Series). The main script is named `Geometric_Lagrangian_vs_Polyakov_Lagrangian` and is fully executable on Google Colab with no additional setup.

8 Conclusions

We have demonstrated the duality between the geometric Lagrangian $\mathcal{L}_{\text{geo}}$ and the Polyakov Lagrangian for a bosonic string compactified on a circle of radius π :

$$\alpha^{-1} = \ln(\lambda_{\text{max}}) - \pi.$$

The circle is not a circle. It is a string. The fine-structure constant is its projected resonance, stripped of the topological phase π .

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